Section 4
DESCRIPTION OF CONDUCTORS

4.1 CONDUCTOR COMPOSITION

Bare Copper (BC)
Tin Coated Copper (TCC)
Silver Coated Copper (SCC)
Nickel Coated Copper Class 2 – Class 10 (2%–10%) (NCC)
Nickel Coated Copper Class 27 "A" Nickel

4.2 TEMPERATURE RANGE

Table 4-1
TEMPERATURE RANGE

Material	Temperature Range	Reference
Bare Copper, Tin-Coated Copper[1], and Silver-Coated Copper	200°C	ASTM B3, ASTM B33, ASTM B298, UL 1581, CSA C22.2 #210.2-M90
Nickel-Coated Copper[2]	250°C	ASTM B355, UL 1581, CSA C22.2 #210.2-M90
Class 27 Nickel-Coated Copper	450°C	ASTM B355, UL 1581, CSA C22.2 #210.2-M90
"A" Nickel	450°C	ASTM B160, UL 1581, CSA C22.2 #210.2-M90

NOTES—
[1] With strand size equal to or larger than 0.015 in.
[2] Nickel Coated Copper Class 2, Class 10 (2%–10%) (NCC)

4.3 GAUGE SIZE

This guideline covers sizes #24 through #8 AWG.

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Table 4-2
TYPICAL WIRE TYPES

Wire Temperature Rating	Conductor Composition*	Insulation Material(s)	Generic Designation
200°C	BC, TCC, NCC Class 2-10 SCC	FEP	FEP
		ETFE	ETFE
		PTFE	PTFE
		Unbraided Silicone Rubber	BSR
		Silicone Rubber/Glass Braid	SF Types
		Polyimide/FEP Tape	KF/KFF
250°C	NCC Class 2 - 10	PFA	PFA
		MFA	MFA
		PTFE	PTFE
		Unbraided Silicone rubber	BSR
		Silicone Rubber/Glass Braid	SF Types
		PTFE/Glass/Glass Braid	TGGT
		PTFE/Ceramic/Glass Braid	TCGT
		PTFE/Aramid/Glass Braid	TKGT
		Polyimide/PTFE Tape	----
		Polyimide/Glass Braid	KGT
450°C	NCC Class 27, "A" Nickel	Mica/Glass Braid	MGT
		Mica/Glass Braid /SS Shield	MGT-Shield

* Per paragraph 4.1

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Section 5
AMPACITIES AND CORRECTION FACTORS

5.1 INTRODUCTION

Establishing ampacity ratings is an inexact procedure. These tables should only be used as a starting point when establishing ratings for any given situation. Values may be lesser or greater than those given in the tables because of the influence of installation method, environment, number of conductors, conductor composition and size, ambient temperatures, insulation types, etc. It is recommended that design engineers desiring accurate ampacity data closely study the 1999 National Electrical Code, Articles 310-15 through 310-84. Additional information can be derived from AIEE, Paper Number 57-660: "The Calculation of the Temperature Rise and Load Capability of Cable Systems" by J. H. Neher and M. H. McGrath. This paper was presented to the AIEE general meeting in Montreal, Quebec, Canada on June 24-28, 1957, and was published in the "AIEE Transactions," Part 3 (power apparatus and systems), Volume 76, October, 1957, pp. 752-772. The information contained therein is still applicable.

The following tables are to be used in series to determine a wire's ampacity in a given application. Reference Section 6 for examples of applications of these tables.

5.2 BASE AMPACITIES

Table 5-1
ALLOWABLE AMPACITIES OF INSULATED CONDUCTORS RATED 0 THROUGH 2000 VOLTS, 60° TO 90°C (140° TO 194° F) NOT MORE THAN THREE CURRENT-CARRYING CONDUCTORS IN RACEWAY OR CABLE OR EARTH (DIRECTLY BURIED), BASED ON AMBIENT TEMPERATURES OF 40°C (104°F)

AWG	150°C[3] BC, TCC	200°C BC, TCC, SCC, or NCC 2%-10%	250°C NCC 2%-10%	250°C "A" Nickel	450°C NCC-Class 27	450°C "A" Nickel
24	6.6	7.2	8	4	9	4.3
22	9	9.6	10.8	5	12	5.6
20	13	14.0	15	7	18	8
18	17	18	20	9.4	23	11
16	22	24	26	12	30	14
14[4]	34	36	39	18	45	21
12	43	45	54	25	56	26
10	55	60	73	34	75	35
8	76	83	93	43	104	49

NOTES—
[3] This column is for reference purposes only.
[4] 14-8 AWG 150-150 NCC data is from NEC Table 310-18.

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Table 5-2
ALLOWABLE AMPACITIES OF INSULATED CONDUCTORS RATED 0 THROUGH 2000 V, 60° TO 90°C (140° TO 194° F) NOT MORE THAN THREE CURRENT-CARRYING CONDUCTORS IN RACEWAY OR CABLE OR EARTH (DIRECTLY BURIED), BASED ON AMBIENT TEMPERATURES OF 40°C (104°F)[5]

AWG	150°C BC, TCC	200°C BC, TCC, SCC, or NCC 2%-10%	250°C NCC 2%-10%	250°C "A" Nickel	450°C NCC-Class 27	450°C "A" Nickel
6[6]	96	110	117	55	138	65
4	120	125	148	69	162	76
3	143	152	166	78	182	85
2	160	171	191	90	210	99
1	186	197	215	101	236	110
1/0	215	229	244	114	268	126
2/0	251	260	273	128	300	141
3/0	288	297	308	144	338	159
4/0	332	346	361	169	397	186
250	365	385	398	187	--	--
300	414	436	452	212	--	--
350	461	486	503	236	--	--
400	495	522	540	254	--	--
500	563	593	613	288	--	--

NOTES—
[5] This table is for reference purposes only.
[6] 6-4/0 150-250 NCC data is from NEC Table 310-18.

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5.3 TEMPERATURE CORRECTION FACTORS

Table 5-3
CORRECTION FACTORS FOR AMBIENT TEMPERATURES OTHER THAN 40°C (104°F), MULTIPLY THE AMPACITIES SHOWN IN TABLES 5-1 AND 5-2 BY THE APPROPRIATE FACTOR SHOWN BELOW

Ambient Temp. C	Wire Temperature Rating 150°C[7]	200°C	250°C	450°C	Ambient Temp. °F
41-50	0.95	0.97	0.98	0.99	106-122
51-60	0.90	0.94	0.95	0.99	124-140
61-70	0.85	0.90	0.93	0.96	142-158
71-80	0.80	0.87	0.90	0.95	160-176
81-90	0.74	0.83	0.87	0.93	177-194
91-100	0.67	0.79	0.85	0.92	195-212
101-120	0.52	0.71	0.79	0.89	213-248
121-140	0.30	0.61	0.72	0.86	249-284
141-160	...	0.50	0.65	0.84	285-320
161-180	...	0.35	0.58	0.81	321-356
181-200	...	...	0.49	0.78	357-392
201-225	...	...	0.35	0.74	393-437
226-250	...	...	...	0.69	439-482
251-275	...	...	...	0.65	483-527
276-300	...	...	...	0.60	528-572
301-325	...	...	...	0.55	573-617
326-350	...	...	...	0.49	618-662
351-375	...	...	...	0.42	663-707
376-400	...	...	...	0.34	708-752
401-450	...	...	...	...	753-842

NOTE—
[7] 150°C data is from NEC Table 310-18.

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5.4 CORRECTION FACTORS FOR MORE THAN THREE CONDUCTORS

Multiply the resulting ampacity calculated from Tables 5-1, 5-2, and 5-3 by the appropriate factors shown below. Since the values in Table 5-4 are expressed as percents, these values are to be converted to the decimal equivalent for calculation purposes.

Table 5-4
ADJUSTMENT FACTORS FOR MORE THAN THREE CURRENT-CARRYING CONDUCTORS IN A RACEWAY OR CABLE
(NEC TABLE 310-15 (b)(2)(a))

Number of Current-Carrying Conductors	Percent of Values in Tables 5-1 through 5-3 as Adjusted for Ambient Temperature if Necessary
4-6	80
7-9	70
10-20	50
21-30	45
31-40	40
41 and above	35

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